Solving for angles in right triangles

1

a 82° **b** 12° **c** 56° **d** 30°

e 45° **f** 50° **g** 61°

2

a $x = 38$ **b** $x = 68$ **c** $y = 47$ **d** $x = 70$

3 greater than 45°

4 $A = 66.4$

5 $t = 47$

6 $x = 53^\circ$

7

a $x = 29.74^\circ$ **b** $y = 150.26^\circ$

8 $A = 49$

9

a 30° **b** $\frac{\sqrt{3}}{2}$ **c** $\frac{1}{\sqrt{3}}$

10

a 30° **b** $\frac{1}{2}$ **c** $\frac{\sqrt{3}}{2}$

11

a i 30° ii $\frac{1}{2}$ **b** i 45° ii $\frac{1}{\sqrt{2}}$ **c** i 30° ii $\frac{1}{2}$

12

a i 60° ii $\sqrt{3}$ **b** i 45° ii 1 **c** i 60° ii $\sqrt{3}$

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13 $\frac{24}{7}$

14

a Yes **b** No **c** No **d** No

15 $\theta = 9.88$

16 $\theta = 54.78^\circ$

17 $\theta = 32.40$

18 $\theta = 49.12$

19 $y = 58.02$

20 $y = 54.90$

21

a $x = 33.31$ **b** $y = 6.97$ **c** $z = 14.39$ **d** 86 cm

22 $x = 22^\circ$

Additional questions

23

a 45° **b** 64° **c** 27° **d** 16°

e 50° **f** 83° **g** 55° **h** 45°

24

a 0° **b** 45° **c** 60° **d** 65°

e 66° **f** 66° **g** 54° **h** 53°

25

a 59.5° **b** 61.2° **c** 83.6° **d** 7.1°

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26

a $x = 74$ **b** $x = 69$ **c** $x = 74^\circ$ **d** $x = 74^\circ$

e $x = 38^\circ$ **f** $x = 53$ **g** $x = 62^\circ$ **h** $x = 67^\circ$

i $x = 67$

27 $x = 21.9^\circ, y = 48.8^\circ$

28

a $x = 26.57^\circ$ **b** $y = 50.19^\circ$ **c** $z = 13.24^\circ$

29

a $\angle 1$ has the greater cosine value. Cosine is a ratio of the adjacent leg to the hypotenuse, and as the angle measure decreases, this ratio gets closer to 1 as the two sides of the triangle almost become the same length.

b $\angle 2$ has the greater sine value. Sine is a ratio of the opposite leg to the hypotenuse, and both triangles have the same opposite leg, so the triangle with the shorter hypotenuse will have the greater value.

c $\angle 2$ has the greater tangent value. Tangent is a ratio of the opposite leg to the adjacent leg, and both triangles have the same opposite leg, so the triangle with the shorter adjacent leg will have the greater value.

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